Fantasy Basketball Roster Analytics

Software Requirements Document

Course: CISC 3171 | Instructor: Alec Malstrom | Term: Fall 2026

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1. Purpose

The Predictive NBA Trade & Roster Evaluator is a web app that acts like a "weather forecast" for fantasy basketball. Instead of looking backward at past stats like Yahoo or ESPN, it predicts how many fantasy points your team will score in the future based on upcoming schedules, team changes, and player injuries.

2. Scope

2.1 In scope

- Evaluating trades of up to 4 players for 4 players.
- Projections for 1-week, 2-week, and rest-of-season (ROS) windows.
- Player search.
- Player filtering and sorting (by position, team, injury status, and projected points).
- Custom league scoring for points leagues and 9-category leagues, with preset scoring options.
- User accounts with saved rosters and saved scoring settings.
- Daily optimal lineup recommendations.
- Ranked add/drop waiver wire recommendations.
- Daily automated stat refreshes with timestamps.

2.2 Out of scope

These are things we are not building, now or later.

- Sports gambling, betting odds, or DFS salary cap tools.
- Making real trades, lineup moves, or waiver claims inside a user's actual league (the app only recommends).
- Native iOS or Android apps (web browser only).
- Notifications for users and direct messaging between users.
- Direct league imports from ESPN, Yahoo, or Sleeper.

3. Context

3.1 User types

We only have two user types, and they are split by what they are allowed to do, not by how often they use the app. Everyone gets the same projections and tools, so a "casual" vs. "competitive" split does not change any functionality. The show/hide stats feature (F8) already covers people who want less detail.

- **Guest:** Not logged in. Can search players, build a roster, change scoring, run trade evaluations, and view projections, lineups, waivers, the glossary, and prediction history. Nothing is saved after the session ends.
- **Registered User:** Has an account. Can do everything a Guest can, plus save rosters and scoring settings.

The word **"user"** in this document means either type. When a feature is limited to one type, the story says Guest or Registered User.

3.2 Operating environment

- Desktop and mobile web browsers, from a 375px phone screen up to desktop.
- The latest two versions of Chrome, Safari, Firefox, and Edge.

3.3 Assumptions and dependencies

- Users enter their rosters manually, since there are no league imports.
- Users know their league's scoring system or use a preset.
- The site depends on an outside NBA data source (stats, schedules, injury reports) that is updated at least once a day. This is why the app has a daily refresh (F6) and a fallback when the source is down.
- Because there are no league imports, the waiver pool is every active NBA player not on the user's roster, minus any players the user marks as taken (see S5.3).

4. Key Features

ID	Feature	Description	Priority	Stories
F1	Roster Building	Search, filter, and sort players; add and remove players on a roster	P1	7
F2	Trade Evaluator	Input trades, calculate net FP or categories, view verdict and top factors	P1	4
F3	Projection Engine	Calculate schedule difficulties, back-to-backs, and injuries	P1	5
F4	Daily Lineup	Show daily projections and recommend starting lineups	P1	2
F5	Data Refresh	Daily updates, timestamp display, and error fallback	P1	2
F6	Tooltips & Glossary	Description of each statistic and how it should affect your decision	P2	2
F7	Stats Filter	Lets users show or hide statistics they are not interested in	P2	2
F8	Custom Scoring	Choose a scoring type, start from a preset, edit weights or categories, save scoring profiles	P1	5

ID	Feature	Description	Priority	Stories
F9	User Accounts	Create an account, log in, and save rosters	P1	5
F10	Prediction History	Record predictions and compare them to actual results	P2	4

Total: 38 user stories.

5. User Stories

F1: Roster Building

S1.1: Searching by name As a user, I want to search for players by name, so I can find the players on my team.

- Must correctly type in the player's name.
- Only active NBA players show up.
- Each result shows team, position eligibility, injury status, and projected FP.

S1.2: Filtering players As a user, I want to filter the player list, so I can narrow down to the players I care about.

- Can filter by position, team, injury status, and whether they are in the waiver wire, is a free agent, or in the user's or other manager's roster.
- Multiple filters can be used at once, and results must match all of them.
- A "Clear filters" button resets the list.

S1.3: Sorting players As a user, I want to sort the player list, so I can see the best options first.

- Can sort by projected FP (for the selected window), name, team, or any visible stat.
- Sort direction can be flipped between high-to-low and low-to-high.
- Default sort is projected FP, high to low.

S1.4: Adding and removing roster players As a user, I want to add and remove players on my roster, so my roster matches my real team.

> **Given:** There are empty slots available to fill.
> **When:** I look at the list of available players.
> **Then:** I am able to add one of the available players to a slot by pressing a button or dragging them into a slot.

> **Given:** There are empty slots available to fill.
> **When:** I look up a player already in the roster.
> **Then:** The player will be grayed out, have a message saying they are already in the roster, and a button to remove the player from the roster.

> **Given:** There are empty slots available to fill.
> **When:** I look up a player already in another manager's roster or is already taken
> **Then:** The player will be grayed out and have a message saying which manager has the player in their roster or is taken, and a button that allows the user to remove the player from that manager's roster and mark them as available.

Given: The roster is full.
When: I look at the list of available players.
Then: The list of available players will be grayed out, cannot be interacted with, and have a message saying your roster is full**.**

Given: I want to remove a player from the roster**.**
When: I click the button to remove a player.
Then: A confirmation box appears asking if I want to remove the player.

Given: I want to add a player that would make it not possible to have a traditional team setup. (e.g. no one in the roster that can play Center)
When: I attempt to add the player to my roster.
Then: A warning appears telling me what positions my team is missing and asking if I want to continue adding the player.

Given: There are players that are unable to be added to a roster.
When: I toggle the "Show/Hide players in rosters" button.
Then: All players that cannot be added will be shown or removed from view.

S1.5: Adding and removing roster players of other managers As a user, I want to track what players are already on another manager's team.

- Separate page with the option to add and remove other managers in the league.
- Able to name the managers.
- Ability to input their current rosters.
- Players the other managers have in their team will be marked.

S1.6: Marking players as taken As a user, I do not want to bother setting up rosters of other managers and just want to mark them as taken or unavailable.

- Players will have individual toggles to mark them as taken.
- Batch option allowing the user to select multiple players to mark them

S1.7: Add/drop suggestions As a user, I want a method to see recommendations that may improve my team.

- Ranks add/drop pairs by projected 7-day FP gain with a 1-line explanation.

F2: Trade Evaluator

S2.1: Building a trade As a user, I want to build a trade proposal, so I can see how it changes my team.

- Supports up to 4 players on each side.
- Players I give come from my roster. Players I receive are found with the player search.
- For uneven trades, the best available waiver pickup is added for each open spot.

S2.2: Points league verdict As a user in a points league, I want a clear verdict, so I know whether to accept.

- Shows Helps Your Team, Hurts Your Team, or Roughly Even.
- Differences within ±5 FP over the selected window are labeled Roughly Even.
- Shows an icon plus a label (never color alone) and the top 3 reasons, each marked + or -.

S2.3: Category league verdict As a user in a 9-category league, I want a verdict based on categories, so it matches how my league is won.

- The verdict line shows the net category change (e.g., "Helps Your Team: +2 categories").
- Below it, a category-by-category list shows each category as better, worse, or tied.
- Helps Your Team if net is +1 or more, Hurts Your Team if net is -1 or less, Roughly Even if net is 0.
- Two categories are tied if they are within 2% of each other (0.5 percentage points for FG% and FT%).

S2.4: Confidence and details As a user, I want to see how sure the app is, so I know how much to trust the verdict.

- Shows a likely range (e.g., +6 to +29 FP) and a confidence level (Low/Med/High).
- A breakdown chart shows each player's contribution.
- Range and breakdown sit in an expandable "Details" section so the first view stays simple.

F3: Projection Engine

S3.1: Changing windows As a user, I want 1-week, 2-week, and ROS windows, so I can plan ahead.

- Toggling the window recalculates points for that specific schedule.
- The selected window shows on every projection screen.

S3.2: Injury adjustments As a user, I want projections to reflect injuries, so evaluations stay accurate.

- Players marked OUT project to 0 FP for the games they miss.
- Other injury statuses reduce projections using the play probabilities in Appendix A.
- Sidelined teammates boost the remaining players' usage rates.

S3.3: What the projection considers As a user, I want to know what goes into a projection, so I can trust it.

- It also uses strength of schedule, number of games in the window, and back-to-backs.
- A "How this is calculated" link lists these inputs in plain language.

S3.4: Viewing a projection As a user, I want projections shown clearly, even in unusual cases, so I am never confused.

- Shows projected FP (or category totals) for each player and for the whole roster.
- If a team has zero games in the window, it shows "No games in this window."
- If a player is out for the whole window, shows 0 FP with an injury label.

S3.5: Predicting other managers As a user, I want to also know how other managers in my league may perform.

- Predictions for how well they may perform assuming they keep the same roster.

F4: Daily Lineup

S4.1: Viewing daily projections As a user, I want to see daily matchups, so I can pick starters.

- Displays slot (PG, SG, C, etc.), opponent, status, and projected daily FP.
- Players with no game that day are marked "No game."

S4.2: Optimizing the lineup As a user, I want 1-click lineup optimization, so I don't bench points.

- An "Optimize" button swaps players to maximize daily points based on position eligibility.
- The user can undo the optimization.

F5: Data Refresh

S5.1: Data timestamp As a user, I want to see when data was last updated, so I know it is fresh.

- Displays a "Data last updated" timestamp on all projection screens.

S5.2: Offline fallback As a user, I want the app to work during source downtime.

- Failed refreshes display cached data alongside a warning banner.

F6: Tooltips & Glossary

S6.1: Stat descriptions As a new user, I want to know what each stat means, so I understand how it should affect my decisions.

- A short description of the stat and how it affects the score shows up by hovering over it (desktop) or tapping an info button (mobile and keyboard).

S6.2: Glossary page As a new user, I want one page with all the terms, so I can learn fantasy basketball in one place.

- The glossary lists every stat and term used in the app (based on Appendix A).
- Tooltips link to their glossary entry.

F7: Stats Filter

S7.1: Showing and hiding stats As a user, I want to hide stats I don't care about, so I am not overloaded with information.

- A toggle shows or hides each statistic.
- Registered Users' choices are saved. Guests' choices last for the session.

S7.2: Resetting hidden stats As a user, I want to bring back hidden stats, so I can see everything again.

- A "Reset" button restores the default list of stats.

F8: Custom Scoring

S8.1: Choosing a scoring type As a user, I want to pick my league's scoring type, so projections match how my league works.

- Two types are supported: Points and 9-Category.
- Points is the default for new rosters.

S8.2: Starting from a preset As a user, I want to start from a preset, so I don't have to enter everything by hand.

- Presets: Standard Points, Expanded Points, and Standard 9-Category (values in Appendix A).
- Picking a preset instantly recalculates projections.
- "Reset to preset" undoes my edits.

S8.3: Editing point values As a user in a points league, I want to edit each stat's point value, so it matches my league.

- Supported stats: PTS, REB, AST, STL, BLK, TO, 3PM, FGM, FGA, FTM, FTA.
- Values can be decimals from -10 to +10.
- Invalid entries (blank or out of range) show an error and are not applied.
- Valid edits instantly recalculate projections.

S8.4: Choosing categories As a user in a 9-category league, I want to choose which categories count, so it matches my league.

- Categories can be turned on or off from the supported stat list.
- The app warns me if the number of categories is not 9.

S8.5: Saving scoring profiles As a Registered User, I want to save scoring settings, so I can reuse them.

- Can save and name up to 3 scoring profiles.
- Each roster can be linked to one profile.
- Guests can still edit scoring, but it is lost when the session ends.

F9: User Accounts

S9.1: Creating an account As a Guest, I want to create an account, so I can save my work.

- Requires a username, email, and password.
- Usernames and emails must be unique.
- A roster a Guest already built carries over into the new account.

S9.2: Logging in and out As a Registered User, I want to log in and out, so my account stays secure.

- Wrong credentials show a generic error that does not say which part was wrong.
- Sessions expire after 30 minutes of inactivity.

S9.3: Saving rosters As a Registered User, I want to save my rosters, so they are there next time.

- Can save up to 3 rosters, each with a name.
- Saved rosters persist across sessions and devices.
- Can rename and delete saved rosters.

S9.4: Resetting a password As a Registered User, I want to be able to reset my password, so I don't lose access to my account.

- A reset link is emailed and expires after 1 hour.

S9.5: Deleting an account As a Registered User, I want to delete my account, so I control my data.

- Deleting requires typing the password.
- All saved rosters and scoring profiles are deleted.

F10: Prediction History

S10.1: Accuracy summary As a user, I want to see how accurate the predictions have been, so I can trust the app.

- Shows average error (in FP), average bias (over- or under-predicting), and the percent of predictions within ±5 FP.
- Can choose the window (1-week or 2-week). ROS accuracy is shown after the season ends.

S10.2: Browsing past predictions As a user, I want to see individual past predictions, so I can check them myself.

- Each row shows date made, player, window, predicted FP, actual FP, and difference.
- Predictions are saved when made and never changed afterward.
- Predictions whose window has not ended show as "Pending."
- Full records are kept for the current and previous season. Older seasons keep monthly accuracy summaries only.

S10.3: Filtering and sorting history As a user, I want to filter and sort the history, so I can find what I'm looking for.

- Filters: player, team, position, window, date range, and scoring type.
- Can sort by any column. Default is the newest first.
- Results are split into pages of 25 rows.

S10.4: Accuracy trend As a user, I want to see accuracy over time, so I can tell if the app is improving.

- A line chart shows average error by week.

6. Non-Functional Requirements

ID	Requirement	Rule
NFR1	Performance	Trade evaluation results return within 3 seconds for a 4-for-4 trade under normal load.
NFR2	Usability	The Trade Evaluator is reachable within 2 clicks/taps from any page, and the verdict shows at the top of the results without scrolling on a 375px screen.
NFR3	Responsiveness	Mobile-friendly design down to 375px width with no sideways scrolling.
NFR4	Security	Passwords are hashed, HTTPS is forced, and no platform logins (ESPN, Yahoo) are requested. Data access follows the table below.
NFR5	Accessibility: Visual	Verdicts use text + icons, never color alone. Text meets 4.5:1 contrast. Text can be enlarged to 200% without losing content.
NFR6	Data freshness	Daily refresh finishes by 6:00 AM Eastern Time.

NFR4 data access

Data	Guest	Registered User
Projections, glossary, prediction history	View	View
Own roster and scoring (current session)	View/edit	View/edit
Saved rosters and scoring profiles	None	View/edit own only
Email and password	None	Own account only (password never viewable)

No one can see another user's email, password, or scoring profiles.

7. Open Questions

There are no open questions right now. OQ1 from the earlier draft was answered and added to the requirements.

OQ1 (resolved): How should 9-category trade verdicts be summarized? Decision: Show both. The verdict line gives the net category change ("+2 categories"), and a category-by-category breakdown sits below it. This matches how the points verdict works (simple answer first, details underneath) and gives casual users a quick answer and competitive users the full picture.

Decided by: Isaac (need project manager to make a final decision it is covered in S2.3). (Project manager did not make decision will consider this open question resolved)

8. Future Directions

These are things we would like to do later.

- **FD1:** Email notification alerts for player injuries.
- **FD2:** Support for rotisserie (season-long category total) scoring.

9. References

- Software Pitch: Fantasy Basketball Roster Analytics Pitch Document (Group 6): [insert link to pitch document]
- Domain Rules: Appendix A of this document.
- NBA stats (example data source): https://www.nba.com/stats
- Basketball Reference stat glossary: https://www.basketball-reference.com/about/glossary.html

Appendix A: Domain Knowledge and Rules

A.1 What is fantasy basketball?

In fantasy basketball, each user (called a manager) builds a team of real NBA players. When those players play real games, the stats they earned count toward the manager's fantasy team. Managers compete against other managers in their league, usually in a weekly head-to-head matchup. The manager with the highest points wins. Managers can improve their team by trading players with other managers, picking up free agents, and choosing which players start each day.

A.2 Glossary

Term	Meaning
Manager	The position the player of a fantasy sport takes on.
Player	Members of an official NBA basketball team. Not to be confused with the player of a fantasy sport.
Roster	All the players on a manager's team. A full roster has up to 14 players, including a slot for an injured player.
League	The group of managers that are competing with each other. Each league has its own rules.
Starting lineup	The players who are active and earn points on a given day.
Bench	Players on the roster who are not in the starting lineup and their points do not contribute to your team's total.
Waiver wire	Players that may be added to the team but require going through the waiver process. These are players that may have been dropped by managers or have already played a game.
Waiver claim	When managers request a player in the waiver wire to be added to their roster. Multiple managers may make a claim for the same player. The winner is decided by their league's rules.
Free agent	Players that may be added without needing to wait for the results of a waiver claim.
Add/drop	Picking up a waiver wire/free agent and releasing a rostered player to make room.
Trade	Swapping players between two managers.
Fantasy points (FP)	The score a player earns from their stats using the league's scoring weights.
Points league	A league where all stats are turned into one total number of fantasy points.
Category (9-cat) league	A league where managers compete in nine separate stats. Winning more categories wins the matchup.
Head-to-head (H2H)	Each week, a manager plays one other manager.
Position eligibility	The positions a player can fill in the lineup (a player can be PG and SG).
Back-to-back (B2B)	Playing games on two days in a row. Players often get more tired or rest on these.

Term	Meaning
Schedule density	How many games a team plays in a given window. More games means more chances to gain fantasy points.
ROS	Rest of season.
Strength of schedule	How good or bad the upcoming opponents are.
Usage rate	The share of a team's plays a player is involved in while on the court. Goes up when key teammates are out.
Confidence level	How sure the app is about a projection (Low, Med, High).

A.3 Stat abbreviations

Abbr.	Stat
PTS	Points scored
REB	Rebounds (missed shots grabbed)
AST	Assists (passes that lead to a basket)
STL	Steals
BLK	Blocks
TO	Turnovers (losing the ball; bad for the manager)
3PM	Three-pointers made
FGM / FGA	Field goals made / attempted
FG%	Field goal percentage (FGM divided by FGA)
FTM / FTA	Free throws made / attempted
FT%	Free throw percentage (FTM divided by FTA)
MIN	Minutes played

A.4 Scoring presets

These are the presets the app offers. Users can edit any of them.

Standard Points:

- PTS +1.0
- REB +1.2
- AST +1.5
- STL +3.0
- BLK +3.0
- TO -1.0.

Expanded Points: Standard Points plus

- 3PM +1.0
- each missed field goal -0.5.

Standard 9-Category: PTS, REB, AST, STL, BLK, 3PM, FG%, FT%, TO. The team with the higher total wins each category. For TO, the lower total wins.

A.5 Positions and lineup slots

- Positions:
    - Point Guard (PG)
    - Shooting Guard (SG)
    - Small Forward (SF)
    - Power Forward (PF)
    - Center (C)
- Variable slots:
    - G can be filled by PG or SG
    - F can be filled by SF or PF
    - UTIL can be filled by any position
- Default lineup (based on common league settings):
    - PG
    - SG
    - SF
    - PF
    - C
    - G
    - F
    - UTL
    - UTL
    - UTL
    - 3 bench spots
    - Injured reserve

A.6 Injury statuses

The app uses these assumed chances of playing. These are our own design assumptions and can be tuned later.

Status	Chance of playing
Available	100%
Probable	75%
Questionable	50%
Doubtful	25%
Out	0%

A.7 How a projection is calculated (simplified)

1. Start with the player's per-36-minute production and scale it to their expected minutes.
2. Adjust for the opponent (strength of schedule), back-to-backs, recent trend, and teammate injuries (usage).
3. Multiply by the chance of playing from A.6.
4. Add up every game in the selected window.

Working example (Standard Points): A player averages 25 PTS, 8 REB, 6 AST, 1 STL, 1 BLK, and 3 TO in a game.

FP = 25 + (8 × 1.2) + (6 × 1.5) + (1 × 3.0) + (1 × 3.0) - 3 = 25 + 9.6 + 9 + 3 + 3 - 3 = **46.6 FP**

With 4 games in a week at this rate, the projection is about 186 FP.

A.8 Example trade verdicts

Points league: A user gives up players projected for 186 FP this week and receives players projected for 200 FP. The net is +14 FP, which is outside the ±5 range, so the verdict is Helps Your Team.

9-category league: A trade improves PTS, REB, AST, STL, and BLK, hurts 3PM, FG%, and FT%, and TO is tied. The net is +2 categories (5 better, 3 worse), so the verdict is Helps Your Team: +2 categories.